

Mutual simulation of Hamiltonian dynamics on interacting quantum systems

Pawel Wocjan, Dominik Janzing, Martin Rötteler, Thomas Beth

Institut für Algorithmen und Kognitive Systeme

Universität Karlsruhe

Control-theoretical model of quantum computers

Time evolution defined by the time-dependent Schrödinger equation

$$\frac{d}{dt}U(t) = -iH(t)U(t), \quad U(0) = \mathbf{1}$$

$H(t)$ Hamiltonian

$U(t)$ is the resulting unitary after time t

$$H(t) = H_d + \sum_{i=1}^m a_i(t)H_i$$

H_d drift Hamiltonian internal to the system

$\sum_{i=1}^m a_i(t)H_i$ control Hamiltonian externally changeable

Complexity of a unitary U : minimal time required to implement U

Fast control limit

strength of the control Hamiltonians arbitrarily large $\implies$

control unitaries in control group K arbitrarily fast

Complexity of a unitary U : minimal time

$$U = \exp(-iHt_N)V_N \cdots \exp(-iHt_2)V_2 \exp(-iHt_1)V_1$$

express it as the evolution according to the piecewise constant Hamiltonian

$$U = U_N \exp(-iU_N^\dagger H U_N t_N) \cdots \exp(-iU_2^\dagger H U_2 t_2) \exp(-iU_1^\dagger H U_1 t_1)$$

Average Hamiltonian Theory

Dyson expansion

$$U(t) = \mathcal{T} \exp\left(\int_{\tau=0}^t d\tau H(\tau)\right) = \exp(-i\bar{H}t)$$

Magnus expansion $\bar{H} = \bar{H}^{(0)} + \bar{H}^{(1)} + \dots$

$$\bar{H}^{(0)} = \frac{1}{t} \int_0^t d\tau H(\tau), \quad \|\bar{H}^{(0)}\| \leq 1$$

$$\bar{H}^{(1)} = \frac{-i}{2t} \int_0^t d\tau' \int_0^t \tau'' [H(\tau'), H(\tau'')], \quad \|\bar{H}^{(1)}\| \leq t/2$$

$\implies U(t)$ essentially determined by $\bar{H}^{(0)}$ for small t

Simulating Hamiltonians (first order simulation)

H can simulate $\tilde{H}$ with overhead 1, written $\tilde{H} \leq H$, and N time steps iff

$$\tilde{H} = \sum_{j=1}^N p_j U_j^\dagger H U_j ,$$

where $p_1, \dots, p_N$ are positive real numbers summing to 1 and $U_j \in K$ are control operations, i.e. $\tilde{H}$ is a convex sum of unitary conjugates of H

H can simulate $\tilde{H}$ with overhead τ iff $\tilde{H} \leq \tau H$

time-optimal simulation problem $\longrightarrow$ convex optimization problem

Composite systems

system consisting of n qudits $\mathcal{H} = \mathbb{C}^d \otimes \mathbb{C}^d \otimes \dots \otimes \mathbb{C}^d$

pair-interaction Hamiltonian

$$H = \sum_{k < l} \sum_{\alpha \beta} J_{kl; \alpha \beta} \sigma_{\alpha}^{(k)} \sigma_{\beta}^{(l)} + \sum_k \sum_{\alpha} r_{k; \alpha} \sigma_{\alpha}^{(k)},$$

where $B := \{\sigma_{\alpha} \mid \alpha = 1, \dots, d^2 - 1\}$ is a basis of $su(d)$

control operations are all local operations in

$$K := SU(d) \otimes SU(d) \otimes \dots \otimes SU(d)$$

Decoupling/Selective decoupling

quantum computing requires the ability to perform gates

couplings created by gates can only originate from natural couplings

naturally available Hamiltonians do not couple specific pairs of qubits

→ turn off disturbing couplings

Time-reversal

evolution of a single spin seems is phenomenologically a relaxation/
dephasing process

however, it is not always irreversible

→ refocusing simulate $-H$ when H is the system Hamiltonian

Universal simulation of quantum dynamics

simulating quantum dynamics on a classical computer is hard

→ universal quantum simulator

use a quantum computer to simulate arbitrary dynamics

Mathematical tools

Vector majorization

let $x = (x_1, x_2, \dots, x_d) \in \mathbb{R}^d$

define $x^\downarrow = (x_1^\downarrow, x_2^\downarrow, \dots, x_d^\downarrow) \in \mathbb{R}^d$ as the vector whose components are the same as of x , but ordered so that

$$x_1^\downarrow \geq x_2^\downarrow \geq \dots \geq x_d^\downarrow$$

x majorizes y , written $x \preceq y$, if

$$\sum_{j=1}^k x_j^\downarrow \leq \sum_{j=1}^k y_j^\downarrow$$

for $k = 1, \dots, d-1$ and

$$\sum_{j=1}^d x_j^\downarrow = \sum_{j=1}^d y_j^\downarrow$$

Mathematical tools

Operator majorization

let A and B be two Hermitian matrices of size d

we say A majorizes B , written $A \preceq B$, if

$$\lambda(A) \preceq \lambda(B),$$

where $\lambda(A)$ and $\lambda(B)$ are the spectra (the vectors of eigenvalues) of A and B , respectively.

Uhlmann's theorem

we have $A \preceq B$ iff

$$A = \sum_{j=1}^N p_j U_j^\dagger B U_j,$$

where $\{p_j\}$ is a probability distribution and $\{U_j\}$ is a set of unitaries